

ATPL EASA STUDY GUIDE | 062 - RADIO NAVIGATION

Chapter 6: Microwave Landing System (MLS) & GLS/SBAS

Syllabus Source: CAE Oxford (pp. 217-252) | Pass Target: $\geq 90\%$ AviationExam

1. MICROWAVE LANDING SYSTEM (MLS)

MLS was developed to overcome ILS frequency saturation and multipath limitations. It operates in the SHF (C-band) from 5030 to 5091 MHz (200 channels):

- Time-Referenced Scanning Beam (TRSB): A narrow beam sweeps back-and-forth across the coverage sector (TO and FRO scan). The time interval between the TO and FRO beam pulses received by the aircraft directly indicates azimuth and elevation angles!
- Operational Capabilities: Azimuth coverage: $\pm 40^\circ$ (up to $\pm 60^\circ$); Elevation: 0.9° to 15° ; Range: 20 NM. Enables CURVED AND SEGMENTED APPROACHES for noise abatement and terrain clearance.

2. SATELLITE-BASED LANDING SYSTEMS (GLS / GBAS)

The Ground-Based Augmentation System (GBAS) provides differential GPS corrections enabling GLS (GBAS Landing System) Cat I, II, and III precision approaches:

- How GLS Works: 4 reference GPS receivers on the airport measure satellite pseudo-range errors. A ground station calculates corrections and broadcasts them along with Final Approach Segment Data Blocks (FASB) over a VHF Data Broadcast (VDB, 108-118 MHz).
- Pilot Interface: The pilot selects a 5-digit channel number on the nav control panel. The cockpit display is identical to ILS, but with pinpoint differential accuracy and zero beam scalloping!

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In which frequency band does the Microwave Landing System (MLS) operate?

[A] UHF [B] SHF (C-Band: 5030 MHz to 5091 MHz) [C] VHF [D] EHF

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. MLS operates in the SHF C-band, rendering it immune to low-frequency site reflections and FM radio interference.

Q2: What is the primary operational advantage of MLS over conventional ILS?

[A] Lower cost [B] Wide-angle proportional coverage permitting CURVED AND SEGMENTED APPROACH PATHS with variable glide slope angles [C] Operates without electricity [D] Long-range en route guidance

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. MLS provides wide azimuth ($\pm 40^\circ$) and elevation coverage, allowing aircraft to fly curved noise-abatement profiles down to touchdown.

Q3: In a GBAS Landing System (GLS), how are the ground differential corrections transmitted to the aircraft?

[A] Via satellite phone [B] Via a VHF DATA BROADCAST (VDB) in the 108.000 to 117.975 MHz band [C] Via HF sky wave [D] Via primary radar

[*] OFFICIAL DEFINITION - Detailed Answer Explanation

CORRECT: [B]. The GBAS ground station broadcasts corrections via an omnidirectional VHF Data Broadcast antenna at the airport.